Examining the effect of face accessories in VLM face evaluation

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Abstract

Vision language models(VLMs) are increasingly used to describe and scan people from a photos. The accessories such as glasses, caps and face masks have no information about a person's age, gender, ethnicity or ability to do a job, yet they hide facial signs and carry social stereotypes. We measure how such accessories change the **VLM** outputs with counterfactual design: 720 synthetic identities, balanced over 4 ethnicities, two genders and thress ages, each shown without an accessory and with glasses, a baseball cap or a face mask added by image editing(2880 images). We query a generative VLM (MiniCPM-V 4.6) and a contrastive image-text model (SigLIP 2) on attribute recognition and job fitness, comapring each accessory image with the same identity's base image using exact paired test. All 18 accessory effects on accuracy are negative and 13 are significant; masks lower age accuracy by about 17 percentage points in both models. The errors have a direction: caps push women towards the label "male" (10-13% vs about 1%) and masks make face look younger. The effects focus on groups that are already harder for the models; e.g., ethnicity errors under caps and masks fall almost entirely on Indian identities. For job fitness, SigLIP 2 shifts along accessory-occupation sterotypes for nearly every identity: a mask raises the "nurse" score of all 720 identities and a cap the "construction worker" score of 716. MiniCPM rates almost everyone as very suitable but still shows a consistent glasses software engineer shift. Accessory counterfactuals should be part of fairness audits of VLMs.

1 Related Work

Face perception in vision-language models (VLMs) has been studied along several largely separate lines, including facial-attribute recognition, social-bias audits, and occlusion-strong facial analysis (Bordes et al., 2024).

1.1 Facial-attribute recognition.

AlDahoul et al. (2024) evaluate GPT-4o, Gemini, LLaVA-NeXT, Florence-2 and PaliGemma on race, gender, age and emotion recognition, and FaceBench (Wang et al., 2025) covers 211 facial attributes across five views, including an Accessories view with thirteen categories. These studies show that VLMs can recognise attributes and accessories, but do not test whether an accessory changes the evaluation of other attributes.

1.2 Occlusion in traditional facial analysis.

Min et al. (2011) detect occluded components with Gabor wavelets, PCA and an SVM and restrict recognition to visible regions with LBP, and Qiu et al. (2022) learn to mask corrupted features inside a deep face-recognition network. The effect extends beyond identity: Yu and Zhao (2025) recover age estimates under masks and sunglasses with knowledge distillation and layer-wise feature reconstruction. These works consider specialised face-analysis models rather than VLMs.

1.3 Controlled face studies and demographic effects.

Hausladen et al. (2025) independently vary age, gender, race, expression, lighting and pose to study CLIP-based social perception, and counterfactual images have been used to isolate demographic effects (Howard et al., 2025; Chen et al., 2026). Hazirbas et al. (2024) relate demographics to eyewear, masks and head coverings, but observationally, and MMBias treats religious and cultural coverings only as coarse

categories (Janghorbani and de Melo, 2023). In humans, masks impair expression recognition most for mouth-dependent emotions (Noyes et al., 2021), and superimposed masks underestimate the effect of real ones (Ritchie et al., 2024), a caveat for any synthetic accessory.

1.4 Research gap.

Prior work studies either occlusion in specialised face-analysis systems or face evaluation with VLMs. VLM benchmarks test whether an accessory is present (Wang et al., 2025), and controlled studies measure the effect of demographic or facial changes (Hausladen et al., 2025; Howard et al., 2025; Chen et al., 2026), but whether accessories systematically change VLM face evaluation remains underexplored. We keep the face and prompt fixed, vary only the accessory, and test whether the evaluation changes and whether the effect differs across demographic groups.

2 Methodology

We investigate whether facial accessories systematically alter how vision-language models (VLMs) evaluate faces and whether these effects vary across demographic groups. We use a controlled within-identity paired-image design in which each original portrait is compared with edited versions of the same portrait containing a target accessory. Within each pair, the identity, initial pose, clothing, and background originate from the same source image; the accessory condition is the intended manipulation. The evaluation prompt, model, and inference settings remain fixed within each pair. This design reduces confounding from differences between identities and allows within-pair changes in model outputs to be associated more directly with the accessory condition.

We examine changes in model predictions. The analysis considers demographic attributes available in the dataset metadata, including age, gender, and ethnicity labels. We also examine subjective, socially loaded judgments,the preceived fit of the portrait for an occupation (job fitness). Finally, we test whether each accessory effects differ across the demographic groups represented in the evaluation sample.

2.1 Research Questions

RQ1. Does adding a facial accessory change VLM outputs for demographic and facial-attribute questions?

We examine whether the addition of a mask, baseball cap, or glasses changes model predictions for attributes represented in the dataset metadata, including age and gender. Because the original and edited images describe the same source identity, differences in model output can be measured within identity.

RQ2. Does adding a facial accessory shift a subjective judgment of the person?

We examine job fitness: how suitable the model judges the person in the portrait to be for three occupations (software engineer, nurse and construction worker), each with a plausible stereotype link to one accessory (glasses, mask and cap). There is no correct answer, so we measure only the shift caused by the accessory, and whether it favours the occupation linked to that accessory.

RQ3. Does the effect of an accessory vary across demographic groups?

We test whether the within-identity change induced by accessory presence differs across the dataset-provided ethnicity labels (African, Asian, Caucasian, and Indian), gender labels (Female and Male), and age values (25, 40, and 55). The focus is not only on differences in absolute VLM scores between demographic groups, but also on whether the effect of an accessory is moderated by demographic group.

2.2 Dataset and Pair Construction

The experimental dataset is a synthetic face dataset generated under controlled conditions (frontal view, near-neutral expression, black t-shirt) (Fournier-Montgieux et al., 2025), containing 9,000 metadata records and 9,000 matching original portrait images. Each record includes a unique image identifier (`HASH`) and dataset-provided labels for ethnicity, gender, and age. The analysis uses only images with matching metadata records. Although the source image directory contains additional files, 3,393 files do not correspond to an identifier in the metadata and are excluded because they cannot be assigned to a demographic cell.

The original dataset is balanced across 24 demographic cells:

$$4\,\text{ethnicity labels} \times 2\,\text{gender labels} \times 3\,\text{age values} = 24\,\text{cells}.$$

Each cell contains 375 portraits, giving:

$$24 \times 375 = 9{,}000 \text{ original portraits.}$$

The labels are African, Asian, Caucasian and Indian; male and female; and 25, 40 and 55.

To make generation and human evaluation feasible while preserving demographic balance, we construct a stratified sample of 720 identities. We select 30 portraits from every ethnicity-gender-age cell:

$$24 \times 30 = 720.$$

Thus, each demographic cell contributes the same number of original portraits to the main evaluation set. The sampled set represents 8% of the full dataset within each cell:

$$\frac{30}{375} = 0.08.$$

Some images in the original dataset already contain accessories, mainly glasses and headwear (e.g., hijab). Glasses Detector (Birškus, 2024) flagged 18.9% of the initial candidate pool (204 of 1,080 images, sampled at 45 per cell) as wearing glasses. Flagged images were removed, and candidates were oversampled so that every cell still ended with exactly 30 identities. Headwear was not filtered: in a manual inspection of 25 random images from the final sample, 2 (8%) showed a head covering. These images were kept, since a cap can be added on top of the headwear or in its place.

For each sampled identity, we create three generated accessory conditions: *mask*, *baseball cap*, and *glasses*. The resulting target set contains:

$$720 \times 3 = 2{,}160 \text{ generated accessory images.}$$

We selected accessories that are common and clearly visible in the image, which excludes items such as earrings or lipstick. Edits were generated with FLUX.2 [klein] 4B (Black Forest Labs, 2026) using the following prompts:

- **Mask:** *"Add a standard white surgical mask covering the lower half of the face. Keeping everything else the same."*
- **Cap:** *"Add a cap. Keep everything else the same."*
- **Eyeglasses:** *"Add eyeglasses. Keep everything else the same."*

Original images are 256×256 px and generated images are 512×512 px. Generation ran on a single A100 GPU at roughly 23 s per image, about 4.7 hours per accessory. The final extended dataset contains 2,880 images (720 originals and 2,160 edited), i.e., 120 images per cell (30 originals and 90 edited). Each generated image is paired with the original portrait from which it was created. The unit of analysis is therefore:

$$(I_i^0, I_i^a),$$

where I_i^0 denotes the original portrait for identity i and I_i^a denotes the corresponding generated portrait with accessory condition $a \in \{\text{mask}, \text{cap}, \text{glasses}\}$.

Both images receive the same prompt, model and inference settings, so every effect is measured within the same portrait and person-specific factors cancel out.

We also retain a separate pilot dataset for development and qualitative inspection. The pilot contains eight identities with five versions each: original, no-edit, mask, cap, and glasses, for a total of 40 files. The pilot set is used to test the generation workflow, refine prompts and review instructions, and select illustrative examples. It is excluded from the main 720-identity evaluation and is not used for demographic or accessory-level statistical comparisons.

Editing drift. Since only the edited images go through FLUX, a paired difference combines two effects:

$$s_{\text{acc}} - s_{\text{orig}} = \underbrace{(s_{\text{ne}} - s_{\text{orig}})}_{\text{editing drift}} + \underbrace{(s_{\text{acc}} - s_{\text{ne}})}_{\text{accessory effect}}, \quad (1)$$

where s is any model score and s_{ne} is the score of the no-edit image, produced by FLUX with the same settings and an instruction to keep the portrait unchanged. In the pilot (Appendix B), the no-edit image increased the SigLIP 2 job similarity in all 24 portrait–job pairs, by $+6.5$ to $+9.5\times10^{-3}$ on average, although it is visually identical to the original. This drift has the same order of magnitude as some accessory effects. We therefore interpret small shifts against the original with caution and discuss the drift as a limitation.

2.3 Human Review of Generated Images

Generated images are evaluated through structured human review before VLM outcome analyses are interpreted. Each reviewer sees the original portrait and its generated counterpart side by side and completes four checks:

1. Is the intended accessory present?
2. Is its placement natural?
3. Is the portrait otherwise preserved?
4. Are there visual defects?

Each check is answered using *Yes*, *No*, or *Uncertain*. Reviewers then provide an overall decision of *Accept*, *Reject*, or *Uncertain*. Notes are required for rejected and uncertain cases. The image identifier, accessory type, reviewer identifier, timestamp, four check responses, overall decision, and free-text notes are stored in CSV file, which provides a reproducible record of the quality-control process and supports analysis of review coverage, acceptance rate, issue frequency, and review outcomes by accessory and demographic cell. In Table 1 we have the human review results:

Table 1: Accessory acceptance statistics.

Accessory	Reviewed	Accepted	Rate
Mask	303	270	0.891
Cap	303	263	0.868
Glasses	84	84	1.000

For glasses all are accepted, while for mask the problem of some generated images is that the mask straps should go behind the ears, not inside the ears, and for the cap there was two rejection causes: sometimes hair is disappeared, and sometimes the cap is placed under the hair.

2.4 Evaluation Protocol

For every original-generated pair, the same prompt and inference settings are used for both images. For demographic and facial-attribute questions, MiniCPM-V-4.6 receives fixed-choice prompts concerning the available dataset labels (ethnicity, gender and age). Accuracy is calculated against the original metadata labels. Because the generated portrait is intended to preserve the source identity, the original image's metadata serves as the reference label for its generated counterpart.

For job fitness, MiniCPM-V-4.6 rates how well the portrait fits each occupation on a five-point scale from A (very unsuitable) to E (very suitable), converted to 1–5 (Section 3.2).

For SigLIP 2, each attribute option and each occupation is represented by a set of text prompts, and the score is the image–text logit (Section 3.2). For prompt set t, the accessory effect for identity i and accessory a is measured as:

$$\Delta_{i,a,t} = s(I_i^a, t) - s(I_i^0, t), \tag{2}$$

where $s(I, t)$ is the image-text score. Positive values indicate a stronger association between the generated accessory image and descriptor t than between the original image and the same descriptor; negative values indicate a weaker association.

For an outcome Y, the within-identity accessory effect in demographic group g is:

$$\Delta Y_g = Y_{\text{accessory},g} - Y_{\text{original},g}. \tag{3}$$

We test whether this within-identity effect differs across ethnicity, gender and age groups by comparing the per-identity changes between groups (Section 3.3). All tests are paired, so each identity serves as its own control. We report effect sizes and confidence intervals alongside statistical significance. Generative and contrastive model outputs are analysed separately because they produce different outcome types.

3 Experimental Setup

The setup was developed in two stages. An initial setup was tested on a pilot and a preliminary run; what we learned from it led to the final setup.

3.1 Initial setup and preliminary results

Pilot. The pipeline was first tested on 8 portraits outside the sample, each in 5 versions(original, no-edits, glasses, cap, mask). Asked for a JSON rating with an abstention option, MiniCPM-V 2.6 returned only 21 valid ratings out of 120 job questions, none for masked images, and all equal to 3 or 4(Appendix B). we therefore moved to MiniCPM 4.6 and to answer formats that always yield a score.

Manipulation check. Asked to name the visible accessory, MiniCPM-V 4.6 named glasses in 720 of 720 (100%), masks in 717 of 720(99.58%) and caps in 719 of 720(99.86%). The 4 other answers named another accessory; since the prompt forces a single name, they don't indicate a failure to see the edit. The manipulation is thus perceived, and the question is if it changes what the models gives about the person.

Preliminary baseline. With one JSON prompt for all 3 attributes, MiniCPM-V 4.6 reached on 720 originals 99.58% on gender, 76.81% on age and 75.42% on ethnicity. Almost all ethnicity errors concerned Indian identities: only 13 of 180 were labelled Indian and 148 were labelled Asian, which in this dataset denotes East Asian faces; for age, 102 of the 240 40-year-olds were labelled 25. With one question per attribute (final setup), the same model reaches 91.9% on ethnicity (72.8% for Indian identities) and 86.5% on age on the same images. Absolute accuracy thus depends strongly on the prompt format, which supports measuring accessory effects as paired changes under a fixed prompt.

3.2 Final setup

Models. MiniCPM-V-4.6 (`openbmb/MiniCPM-V-4.6`) (Yao et al., 2024) is a small open-weight multi-modal language model. We ran it with Hugging Face Transformers, at most 100 new tokens, `downsample_mode=4x`, `max_slice_nums=36` and a new conversation per query. SigLIP-2 (`google/siglip2-giant-opt-patch16-384`) is the largest model of a family of contrastive encoders trained with a sigmoid loss(Zhai et al., 2023; Tschannen et al., 2025); it returns a logit per image-text pair, an affine function of their cosine similarity, and prompts were lower cased and padded to 64 tokens. Encoders of this family are common vision components of larger VLMs.

Objective track (RQ1). For each image and attribute, MiniCPM received one single-choice question with fixed wording and option order (Appendix A). A strict rule extracted the letter; unclear answers would have been mapped to a letter or to NONE (counted as wrong) by a language-model matcher, but all the objective and subjective answers were single letters, with no refusals. For SigLIP 2, each option x was inserted into the four templates of Hausladen et al. (2025) ("a photo of a(n) {x} person.", "a(n) {x} person.", "this is a(n) {x} person.", "cropped face photo of a(n) {x} person."), and the option with the highest mean logit is the answer.

Subjective track (RQ2). We chose 3 jobs, each with reasonable stereotype linked to one accessory: software engineer(technical work, glasses), nurse(care work, mask) and construction worker(manual work, cap). MiniCPM rated suitability from A(Very unsuitable) to E (very suitable), converted to 1-5. For SigLIP 2, job prompts use the four templates ending in "a person suitable for a job as a(n) {job}." and neutral prompts end in "a person.". The job score of image x for job j is

$$s_j(x) = \frac{1}{4}\sum_{t=1}^{4}\Big[\ell\big(x, T_t(j)\big) - \ell\big(x, T_t(\varnothing)\big)\Big], \quad (4)$$

with ℓ the logit, $T_t(j)$ the t-th job prompt and $T_t(\varnothing)$ the t-th neutral prompt. The subtraction removes any change in how well the image matches every prompt about a person.

3.3 Statistical analysis

All comparisons are paired: the output on x_i^a is compared with that on x_i^0. In the objective track, let b be the number of identities answered correctly on the base image and wrongly with the accessory, and c the reverse. The accuracy change is $\Delta = (c-b)/n$ with $n = 720$, tested with the exact McNemar test (McNemar, 1947); reporting b and c also shows changes that cancel out in Δ. In the subjective track, $d_i = v(x_i^a) - v(x_i^0)$ is the change of rating or job score. We report the mean $\bar{d}$, the numbers $n_\downarrow$ and $n_\uparrow$ of identities with $d_i < 0$ and $d_i > 0$, and the net share $100\,(n_\uparrow - n_\downarrow)/n$, which has the same scale for both models, tested with the exact sign test (ties dropped). Confidence intervals are 95% percentile bootstrap intervals over identities with 5,000 resamples (Efron and Tibshirani, 1993).

For RQ3 we repeat the tests inside each ethnicity, gender and age group (180 to 360 identities). Since a significant effect in one group and a non-significant one in another does not show that the groups differ (Gelman and Stern, 2006),

we test the heterogeneity of the per-identity change across groups with the Kruskal–Wallis test (Kruskal and Wallis, 1952), and apply it to base values to measure disparities without accessories. To test whether an accessory favours its stereotype-linked job j^*, we use the contrast $C_i = d_i(j^*) - \frac{1}{2}\sum_{j \neq j^*} d_i(j)$ with the exact sign test. We do not correct for multiple comparisons; with a Bonferroni correction over the 18 tests of each track, 10 of the 13 significant RQ1 effects and 14 of the 15 significant RQ2 effects remain.

4 Results

4.1 RQ1: accessories lower attribute accuracy

Without accessories, both models recognise gender almost perfectly (99.4% for MiniCPM, 99.6% for SigLIP 2) and ethnicity well (91.9%, 92.2%), while age is harder, specially for SigLIP-2 (86.5%, 55.7%). Table 2 shows the paired effects. All 18 changes are negative and 13 are significant. the largest is the mask on age: −17.2 points for MiniCPM(86.5% to 69.3%) and −17.6 for SigLIP-2 (55.7% to 38.1%). Caps lower gender accuracy by 6.1 and 4.6 points and ethnicity accuracy by 6.3 and 4.3, and masks lower ethnicity accuracy by 6.7 and 4.4. Glasses have the smallest effects (at most 2.9 points), significant only for SigLIP-2 on ethnicity and gender.

For gender and ethnicity the changes are almost one-way. with a cap, 44 identities (MiniCPM) and 33(SigLIP-2) go from a correct to a wrong answer and none the other way, with a cap or a mask, 33 to 52 identities lose a correct ethnicity answer while only 2 to 6 gain one. Age answers change in both directions: a mask turns 184 correct MiniCPM answers wrong but also 60 wrong answers correct.

4.2 The error have a direction

Fig. 1 shows how the answers change. With a cap, 13.3% of women are labelled male by MiniCPM and 10.0% by SigLIP-2, against 1.1% and 0.8% without it, while no man is labelled female with a cap. With a mask, face looks younger: MiniCPM's mean signed age error moves from +1 to −4 years overall and from +1.4 to −9.7 years for 40 years old. The 25 years old, who MiniCPM overestimates without mask (+2.3 years), are judged more accurately with one(+.25), which explains the 60 wrong to correct flips. SigLIP-2 judges 55 years old 12.9 years younger with a mask. It also labels 98.3% of the 25 years old as 40 on base images, which explains its low base accuracy on age. Caps are coded as male, and masks hide the lower half of the face, where several signs of age are located.

Table 2: RQ1: accuracy (%) without and with each accessory ($n = 720$ identities). Δ: change in percentage points. Stars: exact McNemar test, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

	MiniCPM-V 4.6		SigLIP 2	
Accessory	Acc.	Δ	Acc.	Δ
Ethnicity				
none	91.9		92.2	
cap	85.7	−6.3***	87.9	−4.3***
glasses	91.0	−1.0	89.3	−2.9***
mask	85.3	−6.7***	87.8	−4.4***
Gender				
none	99.4		99.6	
cap	93.3	−6.1***	95.0	−4.6***
glasses	98.9	−0.6	98.8	−0.8*
mask	98.6	−0.8	98.5	−1.1**
Age				
none	86.5		55.7	
cap	82.5	−4.0**	50.0	−5.7***
glasses	84.9	−1.7	53.2	−2.5
mask	69.3	−17.2***	38.1	−17.6***

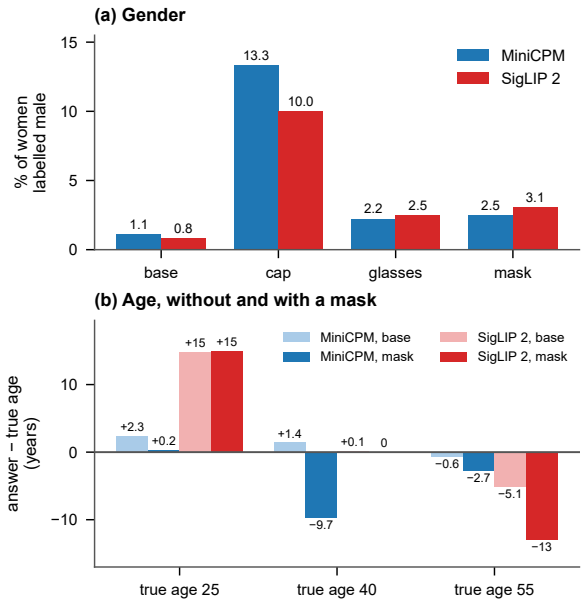


Figure 1: Errors Direction

Table 3: RQ2: effect of each accessory on job fitness ($n = 720$ identities). Δ: mean change in rating (MiniCPM, 1–5 scale) or job score (SigLIP 2, logits, Eq. 4). Net: % of identities rated higher minus % rated lower. †: accessory linked to the job by stereotype. Stars: exact sign test, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

	MiniCPM-V 4.6		SigLIP 2	
Accessory	Δ	Net (%)	Δ	Net (%)
Software engineer				
cap	+0.07	$+6.7^{***}$	+0.24	$+23.6^{***}$
glasses†	+0.11	$+11.0^{***}$	+0.27	$+29.2^{***}$
mask	+0.09	$+8.9^{***}$	−2.16	-96.7^{***}
Nurse				
cap	+0.02	$+2.1^{*}$	+0.79	$+75.8^{***}$
glasses	+0.01	+0.8	+0.14	$+20.0^{***}$
mask†	+0.05	$+4.7^{***}$	+3.32	$+100.0^{***}$
Construction worker				
cap†	+0.05	+1.3	+1.90	$+98.9^{***}$
glasses	+0.03	+0.6	−0.67	-76.1^{***}
mask	−0.15	-3.1^{**}	−0.31	-36.7^{***}

4.3 RQ2: Accessories shift job fitness

MiniCPM rates almost everyone as very suitable: on base images, 87.4%(saoftware engineer), 95.3% and 97.6 % (construction worker) of its answers are E, with mean ratings of 4.87, 4.95 and 4.94. Even with this ceiling, Table 3 shows consistent shifts. Every accessory raises the software engineer rating (net +6.7 to +11.0%, , $p \leq 1.3\times10^{-6}$), the mask and the cap raise the nurse rating (net +4.7%, all 34 changes upwards, and +2.1%), and the mask lowers the construction-worker rating (−0.15 points, net −3.1%, $p = 0.0016$); the three other effects are not significant. All nine SigLIP 2 effects are significant ($p \leq 9.0\times10^{-8}$) and large. A mask raises the nurse score of all 720 identities (+3.32 logits) and lowers the software-engineer score of 708 (−2.16). A cap raises the construction-worker score of 716 identities (+1.90) and the nurse score of 633 (+0.79). Glasses raise the software-engineer and nurse scores of 465 and 432 identities and lower the construction-worker score of 634.

4.4 The shifts follow accessory–occupation stereotypes

For SigLIP 2, the contrast C_i favours the stereotype-linked job for almost every identity: the mask favours "nurse" for all 720 ($\bar{C} = 4.55$ logits), the cap "construction worker" for 713 ($\bar{C} = 1.39$) and glasses "software engineer" for 640 ($\bar{C} = 0.53$; $p < 10^{-100}$ in all three cases). For MiniCPM, only glasses and software engineer show the pattern (107 positive vs. 36 negative contrasts, $p = 2.3\times10^{-9}$); for the mask and nurse the mean contrast is positive but the sign test is not significant (85 vs. 83), and for the cap and construction worker more contrasts are negative (48 vs. 94, $p = 1.4\times10^{-4}$).

These SigLIP 2 results are not an artefact of Eq. 4. Without the neutral subtraction, the raw job logit decreases for seven of the nine accessory–job pairs (−1.0 to −5.2 logits), because an image with an accessory matches "a person" less (the neutral logit drops by 1.8, 2.0 and 3.1 logits for cap, glasses and mask). The only two raw logits that increase are the stereotype-linked pairs mask–nurse (+0.26) and cap–construction worker (+0.12).

4.5 RQ3: some groups are affected much more

Fig. 2 shows up the largest effects by group. Ethnicity errors caused by caps and masks fall almost entirely on Indian identities (−24.4 and −21.7 points for MiniCPM, −14.4 and −14.4 for SigLIP 2), while the other groups changed by −3.3 to +0.6 points. The cap's effect on gender is borne entirely by women (−12.2 and −9.2 points; men 0.0). The mask costs 52.1 points of age accuracy for 40-year-olds in MiniCPM and 55-year-olds in SigLIP 2.

These groups are usually the hardest without the accessories. The basic ethnicity accuracy is 72.8%(MiniCPM) AND 79.4% (sIGlip-2) for Indian identities, against 99.4-100% for Asian and Caucasian ones (Kruskal–Wallis $p = 3.3\times10^{-26}$ and 8.9×10^{-16}), and 40-year-olds have MiniCPM's lowest base age accuracy (79.2%). The accessories widen the existing gaps rather than create new ones. Overall, 22 of the 54 heterogeneity tests are significant at $p < 0.05$, where about three would be expected by chance; the strongest concern age under a mask ($p = 9.5\times10^{-41}$ and 1.8×10^{-63}), ethnicity under a cap (2.5×10^{-23}, 7.2×10^{-12}) and gender under a cap (7.9×10^{-12}, 4.2×10^{-9}).

4.6 RQ3: job fitness depends on gender and age

On base images, SigLIP 2 favours men for construction work (−2.73 vs. −4.18 logits, $p =$

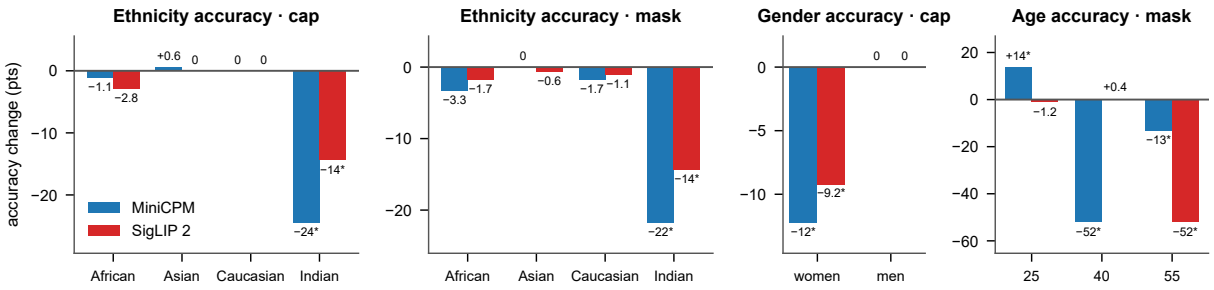


Figure 2: RQ3, objective track: accuracy change inside each group (percentage points) for the largest effects. * exact McNemar $p < 0.05$ inside the group.

$1.4{\times}10^{-66}$) and software engineering (-0.35 vs. -1.20, $p = 3.4{\times}10^{-19}$), women for nursing (-2.81 vs. -3.62, $p = 3.4{\times}10^{-28}$), and younger faces for all jobs, most for software engineering ($+0.01$ at 25 vs. -1.86 at 55; $p = 2.1{\times}10^{-64}$). Despite its ceiling, MiniCPM rates women higher than men for nursing (5.00 vs. 4.91, $p = 2.4{\times}10^{-9}$) and software engineering (4.95 vs. 4.79, $p = 1.7{\times}10^{-10}$), and its software-engineer ratings fall with age (4.96 at 25, 4.78 at 55, $p = 1.1{\times}10^{-8}$).

The accessory effects also differ between groups: 37 of the 54 heterogeneity tests are significant (17 for MiniCPM, 20 for SigLIP 2). For SigLIP 2, a mask lowers construction-worker fitness for men (net -70.0%) but not significantly for women (-3.3%; $p = 1.1{\times}10^{-27}$), and a cap raises software-engineer fitness more for women ($+36.7\%$) than for men ($+10.6\%$, n.s.; $p = 5.2{\times}10^{-6}$). MiniCPM's software-engineer gains are larger for men ($+10.8$ to $+17.8\%$) than for women ($+2.2$ to $+4.2\%$), and for 55-year-olds ($+10.4$ to $+18.3\%$) than for 25-year-olds ($+2.5$ to $+3.8\%$), i.e., for the groups it rated lowest without accessories, which have the most room below the ceiling.

5 Discussion

Accessories are not neutral inputs. In both model families, one everyday accessory changes answers about attributes it does not affect. The effects are large where the accessory hides the evidence a model relies on (masks and age) or carries a social sign (caps and gender), and they are systematic: caps make women look male and masks make people look younger. Applications that infer age or gender from photos can therefore fail predictably for people wearing common accessories.

Some groups are hit harder. The errors fall into specific groups: Indian identities for ethnicity, women for gender, and one age group per model for age. For ethnicity, accessories widen the gap that already exists without them. A fairness audit run only on unoccluded portraits, therefore, underestimates the disparities that occur in practice. Testing heterogeneity directly, rather than comparing which effects per-group are significant (Gelman and Stern, 2006), shows where group differences are real.

Contrastive encoders tie accessories to occupations. SigLIP 2's job scores move almost deterministically along stereotypes. One may argue that these associations are realistic since nurses do wear masks; in a screening setting, however, a feature that says nothing about a person's skills still changes how suitable that person appears. Because such encoders are building blocks of larger VLMs, whether these associations propagate into generative systems deserves study.

High rates can hide bias. MiniCPM rated almost everyone as very suitable, but its ratings still differ by gender and age and rise with glasses for software. Rating-based audits of generative models may need designs without a safe top answer, such as forced choices or pairwise comparisons. Our two setups also show that the answer format matters: MiniCPM-V 2.6 abstained or broke the JSON format in most pilot answers, whereas MiniCPM-V 4.6 answered every forced-choice question, and the same model's ethnicity accuracy rose from 75.4% to 91.9% when attributes were asked separately.

Stability of our findings. The effects that do not survive a Bonferroni correction are small ones: MiniCPM's cap effect on age

($p = 0.0046$), SigLIP 2's glasses and mask effects on gender, and MiniCPM's cap effect on the nurse rating ($p = 0.040$). The numbers of significant heterogeneity tests (22 and 37 of 54) are far above almost three expected by chance.

6 Conclusion

Adding glasses, a cap or a mask to a face changes what VLMs infer about the person, and whom they get wrong. Across 720 counterfactual identities, accessories lowered attribute accuracy in a generative and a contrastive VLM (RQ1), pushed errors in consistent directions, and, for the contrastive model, shifted job-fitness judgments along accessory occupation stereotypes (RQ2). The effects concentrated on specific groups, often those that were already harder for the models (RQ3). Fairness evaluations of VLMs should include accessory counterfactuals and report group heterogeneity explicitly. Future work should cover more models, accessories (e.g., headscarves, hats, beards) and real photographs, add a full-scale no-edit control and a human baseline, and evaluate mitigation.

Limitations

Portraits are synthetic, accessories are generated edits, and edits were reviewed on a sample only; stacked accessories may also underestimate the effect of real ones (Ritchie et al., 2024). Only the edits pass through FLUX: base images are the original 256×256 files and accessory images are 512×512 files, so part of an effect could stem from regeneration (Eq. 1). In the pilot, the no-edit image raised SigLIP 2 job similarity in all 24 portrait–job pairs, by an amount comparable to some accessory effects; measured against it, the glasses effect on "school teacher" changed sign while the large cap and mask effects kept their direction (Appendix B). A full-scale no-edit control would separate the two terms; until then, small effects should be read with care. The demographic labels come from the source dataset and are limited (four ethnicities, binary gender, three ages) and do not reflect self-identification. We used one prompt wording and option order per task; position effects cancel in paired comparisons but affect absolute levels, as the initial setup showed. We evaluated two small open models; larger or proprietary VLMs may behave differently. Finally, judging job fitness from a face is not a valid assessment: we measure it to expose model behaviour that could harm people, not to adopt the task.

Ethics Statement

All faces are synthetic: no real person is depicted and no personal data was processed. The only human involvement was the authors' review of generated images. Demographic labels come with the source dataset and are used only to measure differences; we do not infer religion or any other sensitive attribute beyond the dataset labels. Judging a person's job fitness from a photograph is neither valid nor acceptable as a decision procedure; we deduced such judgments only to show how they can be distorted.

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A Prompts

T0 (MiniCPM-V 4.6). *"Is there an accessory in this image? If yes, answer only with its name: glasses, cap, mask... Otherwise, answer only no."*

T1 (MiniCPM-V 4.6). *"For this image, classify the person using only the following categories. Ethnicity: African, Asian, Caucasian, Indian. Gender: Female, Male. Age: 25, 40, 55. Return JSON only, with exactly these keys: {"ethnicity": "...", "gender": "...", "age": ...}. Choose exactly one value for each attribute from the categories above. Do not provide explanations."*

T2. MiniCPM-V 4.6: *"How suitable does this fictional portrait appear for the role of {job}? Answer with one digit from 1 to 5."* SigLIP 2: *"a portrait of a {job}."* The pilot (MiniCPM-V 2.6) used the same question followed by a request for JSON with a `status` (`rated` or `cannot_assess`) and an integer `score` from 1 to 5 or null.

B Setup Validation Pilot

The measurement chain was tested on 8 portraits outside the evaluation sample, each in five versions (original, no-edit, glasses, cap, mask), i.e. 40 images. SigLIP 2 produced 360 similarities and MiniCPM-V 2.6 answered 160 queries (three job questions and one accessory question per image), without any technical error. The pilot validates the pipeline; it does not test the hypotheses.

	physician	school teacher	delivery driver
No-edit (drift)	+8.3 (8/8)	+9.5 (8/8)	+6.5 (8/8)
Glasses	+14.9 (8/8)	+7.4 (7/8)	−13.7 (7/8)
Cap	−13.7 (8/8)	−18.0 (8/8)	+19.6 (8/8)
Mask	−7.1 (7/8)	−40.0 (8/8)	−37.7 (8/8)

Table 4: Pilot, SigLIP 2, T2: mean paired shift $\bar{\Delta}$ against the original ($\times 10^{-3}$ cosine) and sign consistency k/N ($N = 8$). Original similarities range from 0.056 to 0.125. Descriptive only.

Editing drift. The no-edit row of Table 4 is the first term of Eq. (1). Measured against the no-edit image instead of the original, the glasses effect on *school teacher* changes sign (+7.4 becomes −2.1), while the large cap and mask effects keep their direction.

Generative output coverage. With the JSON prompt, MiniCPM-V 2.6 returned 21 valid ratings out of 120 job questions, 41 abstentions (31 of the 40 *delivery driver* questions) and 58 answers that did not follow the format. No masked image was rated, and all valid ratings were 3 or 4. Abstentions and invalid answers were kept as separate outcomes, never converted to scores. This motivated the forced-choice letter format of the final setup (Section 3.2) and the move to MiniCPM-V 4.6.

C Reproducibility and Compute

Each image is recorded in a manifest with its dataset hash, condition and SHA-256 checksum. Each run stores the pinned model revision, exact prompts, package versions, hardware and a configuration fingerprint, and all raw outputs are kept, including abstentions and invalid answers. In the pilot, MiniCPM-V 2.6 ran on one NVIDIA RTX A5000 (131 s for 160 queries) and SigLIP 2 on four CPU threads (108 s for 40 images), excluding model loading. The full run covers 2,880 images (720 originals and 2,160 edits): 2,880 T1 and 8,640 T2 queries for MiniCPM-V 4.6, and 2,880 image embeddings for SigLIP 2.